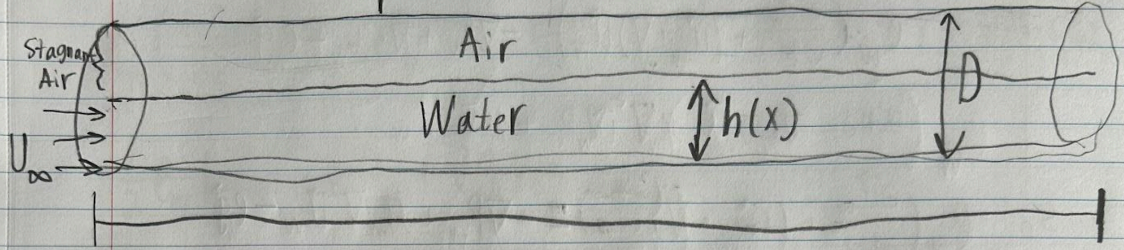
Incompressible Water Pipe Flow



$h(x) < D$ and $\delta_w(x) < h(x)$

$D \ll L$

$\therefore h(x) \ll L$ $\therefore \delta_w(x) \ll L$

Incompressible Continuity (Water)

$$\nabla \cdot \vec{u}_w = 0$$

$$\frac{\partial u_w}{\partial x} + \frac{\partial v_w}{\partial y} = 0$$

Navier-Stokes Momentum (water)

$$\rho_w\left(\frac{\partial \vec{u}_w}{\partial t} + (\vec{u}_w \cdot \nabla)\vec{u}_w\right) = -\nabla P_w + \nabla \cdot \tau_w - \rho\vec{g}$$

$$\tau_w = \mu_w\left(\nabla\vec{u}_w + (\nabla\vec{u}_w)^T\right) + \lambda_w \cancel{(\nabla \cdot \vec{u}_w)}_{0}$$

$$\nabla \cdot \tau_w = \mu_w \nabla^2 \vec{u}_w + \mu\nabla\cancel{(\nabla \cdot \vec{u}_w)}_{0}$$

$$\nabla \cdot \tau_w = \mu_w \nabla^2 \vec{u}_w$$

Incompressible Pipe Flow

Since $\rho_w\left(\frac{\partial \vec{u}_w}{\partial t} + (\vec{u}_w\cdot\nabla)\vec{u}_w\right) = -\nabla P_w + \nabla\cdot\tau_w - \rho_w\vec{g}$

and $\nabla\cdot\tau_w = \mu_w\nabla^2\vec{u}_w$

So $\rho_w\left(\frac{\partial \vec{u}_w}{\partial t} + (\vec{u}_w\cdot\nabla)\vec{u}_w\right) = -\nabla P_w + \mu_w\nabla^2\vec{u}_w - \rho_w\vec{g}$

$$(\vec{u}_w\cdot\nabla)\vec{u}_w = \nabla\left(\tfrac{1}{2}|\vec{u}_w|^2\right) + \vec{\omega}_w\times\vec{u}_w$$

$$\rho_w\left(\frac{\partial \vec{u}_w}{\partial t} + \nabla\left(\tfrac{1}{2}|\vec{u}_w|^2\right) + \vec{\omega}_w\times\vec{u}_w\right) = -\nabla P_w + \mu_w\nabla^2\vec{u}_w - \rho_w\vec{g}$$

$$\rho_w\left(\frac{\partial \vec{u}_w}{\partial t} + \vec{\omega}_w\times\vec{u}_w\right) + \rho_w\nabla\left(\tfrac{1}{2}|\vec{u}_w|^2\right) = -\nabla P_w - \rho_w\vec{g} + \mu_w\nabla^2\vec{u}_w$$

$$\rho_w\vec{g} = \begin{pmatrix}0\\ \rho_w g\end{pmatrix} \qquad \phi_g = \rho_w g y$$

$$\nabla\phi_g = \begin{pmatrix}0\\ \rho_w g\end{pmatrix} = \rho_w\vec{g}$$

$$\nabla(\rho_w g y) = \rho_w\vec{g}$$

$$\rho_w\left(\frac{\partial \vec{u}_w}{\partial t} + \vec{\omega}_w\times\vec{u}_w\right) + \nabla\left(\tfrac{1}{2}\rho_w|\vec{u}_w|^2\right) = -\nabla P_w - \nabla(\rho_w g y) + \mu_w\nabla^2\vec{u}_w$$

$$\rho_w\left(\frac{\partial \vec{u}_w}{\partial t} + \vec{\omega}_w\times\vec{u}_w\right) = -\nabla\left(\tfrac{1}{2}\rho_w|\vec{u}_w|^2\right) - \nabla P_w - \nabla(\rho_w g y) + \mu_w\nabla^2\vec{u}_w$$

Incompressible Pipe Flow

$$\rho_w\left(\frac{\partial \vec{u}_w}{\partial t} + \vec{\omega}_w \times \vec{u}_w\right) = -\nabla\left(\tfrac{1}{2}\rho_w|\vec{u}_w|^2 + P_w + \rho_w g y\right) + \mu_w \nabla^2 \vec{u}_w$$

$$\frac{1}{2}\rho_w|\vec{u}_w|^2 + P_w + \rho_w g y = P_{0,w}$$

$$\rho_w\left(\frac{\partial \vec{u}_w}{\partial t} + \vec{\omega}_w \times \vec{u}_w\right) = -\nabla P_{0,w} + \mu_w \nabla^2 \vec{u}_w$$

$$\nabla \times \vec{u}_w = \vec{\omega}_w$$

$$\vec{\omega}_w = \nabla \times \vec{u}_w = \begin{pmatrix} \partial/\partial x \\ \partial/\partial y \\ 0 \end{pmatrix} \times \begin{pmatrix} u_w \\ v_w \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \frac{\partial v_w}{\partial x} - \frac{\partial u_w}{\partial y} \end{pmatrix} \quad 2D$$

$$\vec{\omega}_w \times \vec{u}_w = \begin{pmatrix} 0 \\ 0 \\ \frac{\partial v_w}{\partial x} - \frac{\partial u_w}{\partial y} \end{pmatrix} \times \begin{pmatrix} u_w \\ v_w \\ 0 \end{pmatrix} = \det\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & \frac{\partial v_w}{\partial x} - \frac{\partial u_w}{\partial y} \\ u_w & v_w & 0 \end{vmatrix}$$

$$\vec{\omega}_w \times \vec{u}_w = \left(\frac{\partial u_w}{\partial y} - \frac{\partial v_w}{\partial x}\right) v_w \hat{i} + \left(\frac{\partial v_w}{\partial x} - \frac{\partial u_w}{\partial y}\right) u_w \hat{j}$$

$$\vec{\omega}_w \times \vec{u}_w = \begin{pmatrix} \left(\frac{\partial u_w}{\partial y} - \frac{\partial v_w}{\partial x}\right) v_w \\ \left(\frac{\partial v_w}{\partial x} - \frac{\partial u_w}{\partial y}\right) u_w \end{pmatrix} \quad 2D$$

Incompressible Pipe Flow

Since

$$\rho_w\left(\frac{\partial \vec{u}_w}{\partial t} + \vec{\omega}_w \times \vec{u}_w\right) = -\nabla P_{0,w} + \mu_w \nabla^2 \vec{u}_w ; \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

And $\vec{\omega}_w \times \vec{u}_w = \begin{pmatrix} \left(\frac{\partial u_w}{\partial y} - \frac{\partial v_w}{\partial x}\right) v_w \\ \left(\frac{\partial v_w}{\partial x} - \frac{\partial u_w}{\partial y}\right) u_w \end{pmatrix}$

X-Momentum (Water)

$$\rho_w\left(\frac{\partial u_w}{\partial t} + \left(\frac{\partial u_w}{\partial y} - \frac{\partial v_w}{\partial x}\right) v_w\right) = -\frac{\partial P_{0,w}}{\partial x} + \mu_w \frac{\partial^2 u_w}{\partial x^2} + \mu_w \frac{\partial^2 u_w}{\partial y^2}$$

Nondimensionalize

$$\frac{\partial u_w}{\partial x} \sim \frac{U_\infty}{L}, \quad \frac{\partial u_w}{\partial y} \sim \frac{U_\infty}{h}, \quad \frac{\partial v_w}{\partial x} = \frac{\bar{v}_w}{L}, \text{ and } \frac{\partial v_w}{\partial y} = \frac{\bar{v}_w}{h}$$

Where $\bar{v}_w$ is the spatial average y-direction velocity of water.

Therefore, the nondimensional variables: $u'_w = \frac{u_w}{U_\infty}$, $v'_w = \frac{v_w}{\bar{v}_w}$, $x' = \frac{x}{L}$, and $y' = \frac{y}{h}$

$$\rho_w\left(\frac{\partial u_w}{\partial t} + \left(\frac{U_\infty}{h}\frac{\partial u'_w}{\partial y'} - \frac{\bar{v}_w}{L}\frac{\partial v'_w}{\partial x'}\right) v_w\right) = -\frac{\partial P_{0,w}}{\partial x} + \mu_w \frac{U_\infty}{L^2}\frac{\partial^2 u'_w}{\partial x'^2} + \mu_w \frac{U_\infty}{h^2}\frac{\partial^2 u'_w}{\partial y'^2}$$

$$\therefore \rho_w\left(\frac{\partial u_w}{\partial t} + \left(\frac{U_\infty}{h}\frac{\partial u'_w}{\partial y'} - \frac{\bar{v}_w}{L}\frac{\partial v'_w}{\partial x'}\right) v_w\right) = -\frac{\partial P_{0,w}}{\partial x} + \mu_w \frac{U_\infty}{h^2}\left(\underbrace{\cancel{\frac{h^2}{L^2}}\frac{\partial^2 u'_w}{\partial x'}}_{0,\ h \ll L} + \frac{\partial^2 u'_w}{\partial y'^2}\right)$$

Incompressible Pipe Flow

So,

$$\rho_w\left(\frac{\partial u_w}{\partial t}+\left(\frac{U_\infty}{h}\frac{\partial u'}{\partial y'}-\frac{\bar{v}_w}{L}\frac{\partial v'_w}{\partial x'}\right)v_w\right)=-\frac{\partial P_{0,w}}{\partial x}+\mu_w\frac{U_\infty}{h^2}\frac{\partial^2 u'_w}{\partial y'^2}$$

Redimensionalize right-side of equation.

$$\therefore \rho_w\left(\frac{\partial u_w}{\partial t}+\left(\frac{U_\infty}{h}\frac{\partial u'}{\partial y'}-\frac{\bar{v}_w}{L}\frac{\partial v'_w}{\partial x'}\right)v_w\right)=-\frac{\partial P_{0,w}}{\partial x}+\mu_w\frac{\partial^2 u_w}{\partial y^2}$$

Incompressible Continuity

$$\frac{\partial u_w}{\partial x}+\frac{\partial v_w}{\partial y}=0$$

Nondimensionalize

$$\frac{U_\infty}{L}\frac{\partial u'_w}{\partial x'}+\frac{\bar{v}_w}{h}\frac{\partial v'_w}{\partial y'}=0$$

$$\frac{h}{U_\infty}\left(\frac{U_\infty}{L}\frac{\partial u'_w}{\partial x'}+\frac{\bar{v}_w}{h}\frac{\partial v'_w}{\partial y'}\right)=\left(\frac{h}{U_\infty}\right)(0)$$

$$\underset{h\ll L}{\cancel{\frac{h}{L}}}\frac{\partial u'_w}{\partial x'}+\frac{\bar{v}_w}{U_\infty}\frac{\partial v'_w}{\partial y'}=0$$

$$\frac{\bar{v}_w}{U_\infty}\frac{\partial v'_w}{\partial y'}=0 \text{ and } \frac{\partial v'_w}{\partial y'}\sim O(1) \therefore \frac{\bar{v}_w}{U_\infty}O(1)\cong 0$$

Incompressible Pipe Flow

Since $\frac{\bar{V}_w}{U_\infty} O(1) \cong 0$, $\frac{\bar{V}_w}{U_\infty} \cong 0$, $\therefore \underbrace{\bar{V}_w << U_\infty}$

$$\bar{V}_w \cong 0$$

And $\rho_w \left(\frac{\partial u_w}{\partial t} + \left(\frac{U_\infty}{h} \frac{\partial u'}{\partial y'} - \frac{\bar{V}_w}{L} \frac{\partial v'_w}{\partial x'} \right) v_w \right) = -\frac{\partial P_{0,w}}{\partial x} + \mu_w \frac{\partial^2 u}{\partial y^2}$

$$\therefore \rho_w \left(\frac{\partial u_w}{\partial t} + \left(\frac{U_\infty}{h} \right) \left(\frac{\partial u'}{\partial y'} - \cancel{\frac{h \bar{V}_w}{L U_\infty} \frac{\partial v'_w}{\partial x'}} \right) v_w \right) = -\frac{\partial P_{0,w}}{\partial x} + \mu_w \frac{\partial^2 u}{\partial y^2}$$

0, $h << L$
and $\bar{V}_w << U_\infty$

$$\rho_w \left(\frac{\partial u_w}{\partial t} + \frac{U_\infty}{h} \frac{\partial u'}{\partial y'} v_w \right) = -\frac{\partial P_{0,w}}{\partial x} + \mu_w \frac{\partial^2 u}{\partial y^2}$$

Redimensionalize left side of the equation

$$\rho_w \left(\frac{\partial u_w}{\partial t} + v_w \frac{\partial u}{\partial y} \right) = -\frac{\partial P_{0,w}}{\partial x} + \mu_w \frac{\partial^2 u}{\partial y^2} \quad ①$$

Incompressible Pipe Flow

Since $\rho_w\left(\frac{\partial \vec{u}_w}{\partial t} + \vec{\omega}_w \times \vec{u}_w\right) = -\nabla P_{0,w} + \mu_w \nabla^2 \vec{u}_w$

And $\vec{\omega}_w \times \vec{u}_w = \begin{pmatrix} \left(\frac{\partial u_w}{\partial y} - \frac{\partial v_w}{\partial x}\right) v_w \\ \left(\frac{\partial v_w}{\partial x} - \frac{\partial u_w}{\partial y}\right) u_w \end{pmatrix}$

Y-momentum (Water)

$$\rho_w\left(\frac{\partial v_w}{\partial t} + \left(\frac{\partial v_w}{\partial x} - \frac{\partial u_w}{\partial y}\right) u_w\right) = -\frac{\partial P_{0,w}}{\partial y} + \mu_w \frac{\partial^2 v_w}{\partial x^2} + \mu_w \frac{\partial^2 v_w}{\partial y^2}$$

Nondimensionalize

$$\rho_w\left(\frac{\partial v_w}{\partial t} + \left(\frac{\bar{v}_w}{L}\frac{\partial v'_w}{\partial x'} - \frac{U_\infty}{h}\frac{\partial u'_w}{\partial y'}\right) u_w\right) = -\frac{\partial P_{0,w}}{\partial y} + \mu_w \frac{\bar{v}_w}{L^2}\frac{\partial^2 v'_w}{\partial x'^2} + \mu_w \frac{\bar{v}_w}{h^2}\frac{\partial^2 v'_w}{\partial y'^2}$$

$$\rho_w\left(\frac{\partial v_w}{\partial t} + \left(\frac{U_\infty}{h}\right)\left(\underbrace{\cancelto{0}{\frac{h\bar{v}_w}{LU_\infty}}}_{\substack{h \ll L \\ \text{and } \bar{v}_w \ll U_\infty}}\frac{\partial v'_w}{\partial x'} - \frac{\partial u'_w}{\partial y'}\right) u_w\right) = -\frac{\partial P_{0,w}}{\partial y} + \mu_w \frac{\bar{v}_w}{h^2}\left(\underbrace{\cancelto{0}{\frac{h^2}{L^2}}}_{h \ll L}\frac{\partial^2 v'_w}{\partial x'^2} + \frac{\partial^2 v'_w}{\partial y'^2}\right)$$

$$\rho_w\left(\frac{\partial v_w}{\partial t} - \frac{U_\infty}{h}\frac{\partial u'_w}{\partial y'} u_w\right) = -\frac{\partial P_{0,w}}{\partial y} + \mu_w \frac{\bar{v}_w}{h^2}\frac{\partial^2 v'_w}{\partial y'^2}$$

Redimensionalize

$$\rho_w\left(\frac{\partial v_w}{\partial t} - u_w \frac{\partial u_w}{\partial y}\right) = -\frac{\partial P_{0,w}}{\partial y} + \mu_w \frac{\partial^2 v_w}{\partial y^2}$$

Incompressible Pipe Flow

$$\rho_w\left(\frac{\partial v_w}{\partial t} - u_w\frac{\partial u_w}{\partial y}\right) = -\frac{\partial P_{0,w}}{\partial y} + \mu_w\frac{\partial^2 v_w}{\partial y^2}$$

$$\rho_w\left(\frac{\partial v_w}{\partial t} - \frac{\partial\left(\frac{1}{2}u_w^2\right)}{\partial y}\right) = -\frac{\partial P_{0,w}}{\partial y} + \mu_w\frac{\partial^2 v_w}{\partial y^2}$$

$$P_{0,w} = P_w + \frac{1}{2}\rho_w|\vec{U}_w|^2 + \rho_w g y$$

$$\therefore P_{0,w} = P_w + \frac{1}{2}\rho_w u_w^2 + \frac{1}{2}\rho_w v_w^2 + \rho_w g y$$

$$\rho_w\frac{\partial v_w}{\partial t} - \rho_w\frac{\partial\left(\frac{1}{2}u_w^2\right)}{\partial y} = -\frac{\partial\left(P_w + \frac{1}{2}\rho_w u_w^2 + \frac{1}{2}\rho_w v_w^2 + \rho_w g y\right)}{\partial y} + \mu\frac{\partial^2 v_w}{\partial y^2}$$

$$\rho_w\frac{\partial v_w}{\partial t} - \cancel{\frac{\partial\left(\frac{1}{2}\rho_w u_w^2\right)}{\partial y}} = -\frac{\partial\left(P_w + \frac{1}{2}\rho_w v_w^2 + \rho_w g y\right)}{\partial y} - \cancel{\frac{\partial\left(\frac{1}{2}\rho_w u_w^2\right)}{\partial y}} + \mu_w\frac{\partial^2 v}{\partial y^2}$$

$$\rho_w\frac{\partial v_w}{\partial t} = -\frac{\partial\left(P_w + \frac{1}{2}\rho_w v_w^2 + \rho_w g y\right)}{\partial y} + \mu_w\frac{\partial^2 v}{\partial y^2}$$

$P_w + \frac{1}{2}\rho_w v_w^2 + \rho_w g y$, nondimensionalize the expression

$$P'_w = \frac{P_w - P_\infty}{\frac{1}{2}\rho_w U_\infty^2}, \quad v'_w = \frac{v_w}{\bar{v}_w} \qquad \therefore \frac{1}{2}\rho_w U_\infty^2 P'_w + P_\infty + \frac{1}{2}\rho_w \bar{v}_w^2 v_w'^2 + \rho_w g y$$

$\Rightarrow P_w = \frac{1}{2}\rho_w U_\infty^2 P'_w + P_\infty$ and $v_w = \bar{v}_w v'_w$

Incompressible Pipe Flow

Approximate the expression

$$\frac{1}{2}\rho_w U_\infty^2 P_w' + P_\infty + \frac{1}{2}\rho_w \overline{V_w}^2 v_w'^2 + \rho_w g y$$

$$\frac{1}{2}\rho_w U_\infty^2 \left(P_w' + \frac{\cancel{\frac{1}{2}\rho_w}\overline{V_w}^2}{\cancel{\frac{1}{2}\rho_w} U_\infty^2}(v_w')^2\right) + P_\infty + \rho_w g y$$

$$\frac{1}{2}\rho_w U_\infty^2 \left(P_w' + \cancel{\frac{\overline{V_w}^2}{U_\infty^2}(v_w')^2}\right] + P_\infty + \rho_w g y$$

$0, \overline{V_w} << U_\infty$

$$\therefore \frac{1}{2}\rho_w U_\infty^2 P_w' + P_\infty + \rho_w g y$$

$P_w + \rho_w g y$ is now the expression

So $$\rho_w \frac{\partial v_w}{\partial t} = -\frac{\partial(P_w + \frac{1}{2}\rho_w \overline{V_w}^2 + \rho_w g y)}{\partial y} + \mu_w \frac{\partial^2 v_w}{\partial y^2}$$

$$\cong \rho_w \frac{\partial v_w}{\partial t} = -\frac{\partial(P_w + \rho_w g y)}{\partial y} + \mu \frac{\partial^2 v_w}{\partial y^2}$$

$$\rho_w \frac{\partial v_w}{\partial t} = -\frac{\partial P_w}{\partial y} - \rho_w g + \mu \frac{\partial^2 v_w}{\partial y^2}$$

$$\rho_w \frac{\partial v_w}{\partial t} + \frac{\partial P_w}{\partial y} + \rho_w g = \mu \frac{\partial^2 v_w}{\partial y^2} \quad ②$$

Incompressible Pipe Flow

① $$\rho_w\left(\frac{\partial u_w}{\partial t} + v_w\frac{\partial u_w}{\partial y}\right) = -\frac{\partial P_{0,w}}{\partial x} + \mu_w\frac{\partial^2 u_w}{\partial y^2}$$

② $$\rho_w\frac{\partial v_w}{\partial t} + \frac{\partial P_w}{\partial y} + \rho_w g = \mu_w\frac{\partial^2 v_w}{\partial y^2}$$

Where $P_{0,w} = \frac{1}{2}\rho_w|\vec{U}_w|^2 + \rho_w g y + P_w$

$$P_{0,w} = \frac{1}{2}\rho_w(u_w^2 + v_w^2) + P_w + \rho_w g y$$

$$\frac{\bar{V}_w}{U_\infty} \sim \frac{h}{L}$$

$$P_{0,w} = \frac{1}{2}\rho_w U_\infty^2\left(u_w'^2 + \cancelto{0,\ \bar{V}_w \ll U_\infty}{\left(\frac{\bar{V}_w}{U_\infty}\right)^2} v_w'^2\right) + P_w + \rho_w g y$$

$$P_{0,w} = \frac{1}{2}\rho_w U_\infty^2 u_w'^2 + P_w + \rho_w g y$$

$$P_{0,w} = \frac{1}{2}\rho_w u_w^2 + P_w + \rho_w g y$$

$$\therefore \frac{\partial P_{0,w}}{\partial x} = \frac{\partial\left(\frac{1}{2}\rho_w u_w^2 + P_w + \rho_w g y\right)}{\partial x} = \rho_w u_w\frac{\partial u_w}{\partial x} + \frac{\partial P_w}{\partial x}$$

$$-\frac{\partial P_{0,w}}{\partial x} = -\rho_w u_w\frac{\partial u_w}{\partial x} - \frac{\partial P_w}{\partial x}$$

Incompressible Pipe Flow

Since,

$$\rho_w\left(\frac{\partial u_w}{\partial t} + v_w \frac{\partial u_w}{\partial y}\right) = -\frac{\partial P_{o,w}}{\partial x} + \mu_w \frac{\partial^2 u_w}{\partial y^2}$$

$$\rho_w\left(\frac{\partial u_w}{\partial t} + v_w \frac{\partial u_w}{\partial y}\right) = -\rho_w U_w \frac{\partial u_w}{\partial x} - \frac{\partial P_w}{\partial x} + \mu_w \frac{\partial^2 u_w}{\partial y^2}$$

$$\rho_w\left(\frac{\partial u_w}{\partial t} + U_w \frac{\partial u_w}{\partial x} + v_w \frac{\partial u_w}{\partial y}\right) = -\frac{\partial P_w}{\partial x} + \mu_w \frac{\partial^2 u_w}{\partial y^2} \quad ①$$

$$\rho_w \frac{\partial v_w}{\partial t} + \frac{\partial P_w}{\partial y} + \rho_w g = \mu_w \frac{\partial^2 v_w}{\partial y^2} \quad ②$$

$$\int_0^y \rho_w \frac{\partial v_w}{\partial t} \partial \bar{y} + \int_0^y \frac{\partial P_w}{\partial \bar{y}} \partial \bar{y} + \int_0^y \rho_w g \, \partial \bar{y} = \int_0^y \mu_w \frac{\partial^2 v_w}{\partial \bar{y}^2} \partial \bar{y}$$

$$\int_0^y \rho_w \frac{\partial v_w}{\partial t} v_w \partial t + P_w - P_w\Big|_{y=0} + \rho_w g y = \mu \frac{\partial v_w}{\partial y} - \underset{0}{\cancel{\mu \frac{\partial v_w}{\partial y}\Big|_{y=0}}}$$

Continuity (Water): $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0,$

At wall, $y=0$

$$\underset{0}{\cancel{\frac{\partial u}{\partial x}\Big|_{y=0}}} + \frac{\partial v}{\partial y}\Big|_{y=0} = 0, \quad u(x, y=0) = 0, \text{ no slip} \quad \frac{\partial v}{\partial y}\Big|_{y=0} = 0$$

$$\frac{\partial}{\partial x}\left(u(x, y=0)\right) = \frac{\partial}{\partial x}(0)$$

$$\frac{\partial u}{\partial x}\Big|_{y=0} = 0$$

Incompressible Pipe Flow

$$\int_0^y \rho_w \frac{\partial v_w}{\partial t} v_w \partial t + P_w - P_w\Big|_{y=0} + \rho_w g y = \mu_w \frac{\partial v}{\partial y}$$

$$\int_0^y \rho_w v_w \frac{\partial v_w}{\partial t} \partial t + P_w - P_w\Big|_{y=0} + \rho_w g y = \mu_w \frac{\partial v_w}{\partial y}$$

$$\int_0^y \rho_w v_w \partial v_w + P_w - P_w\Big|_{y=0} + \rho_w g y = \mu_w \frac{\partial v_w}{\partial y}$$

$$\frac{1}{2}\rho_w v_w^2 - \overset{0}{\cancel{\frac{1}{2}\rho_w v_w^2\Big|_{y=0}}} + P_w - P_w\Big|_{y=0} + \rho_w g y = \mu_w \frac{\partial v_w}{\partial y}$$

$v_w|_{y=0} = 0$, due to no permeability

$$\frac{1}{2}\rho_w v_w^2 + P_w - P_w\Big|_{y=0} + \rho_w g y = \mu_w \frac{\partial v_w}{\partial y}$$

$$P_w' = \frac{P_w - P_\infty}{\frac{1}{2}\rho_w U_\infty^2}, \quad v_w' = \frac{v_w}{\bar{v}_w}$$

$$P_w = \frac{1}{2}\rho_w U_\infty^2 P_w' + P_\infty \quad \text{and} \quad v_w = \bar{v}_w v_w'$$

$$\frac{1}{2}\rho_w \bar{v}_w^2 v_w'^2 + \frac{1}{2}\rho_w U_\infty^2 P_w' + P_\infty - P_w\Big|_{y=0} + \rho_w g y = \mu_w \frac{\partial v}{\partial y}$$

Incompressible Pipe Flow

$$\frac{1}{2}\rho_w \bar{v}_w^2 (v_w')^2 + \frac{1}{2}\rho_w U_\infty^2 P_w' + P_\infty - P_w\Big|_{y=0} + \rho_w g y = \mu_w \frac{\partial v_w}{\partial y}$$

$$\frac{1}{2}\rho_w U_\infty^2 \left(\cancel{\frac{\bar{v}_w^2}{U_\infty^2}(v_w')^2} + P_w' \right) + P_\infty - P_w\Big|_{y=0} + \rho_w g y = \mu_w \frac{\partial v_w}{\partial y}$$

$\bar{v}_w \ll U_\infty$

$$\underbrace{\frac{1}{2}\rho_w U_\infty^2 P_w' + P_\infty}_{P_w} - P_w\Big|_{y=0} + \rho_w g y = \mu_w \frac{\partial v_w}{\partial y}$$

$$\therefore P_w - P_w\Big|_{y=0} + \rho_w g y = \mu_w \frac{\partial v_w}{\partial y} \quad ②$$

$$\rho_w \left(\frac{\partial u_w}{\partial t} + u_w \frac{\partial u_w}{\partial x} + v_w \frac{\partial u_y}{\partial y} \right) = -\frac{\partial P_w}{\partial x} + \mu_w \frac{\partial^2 u_w}{\partial y^2} \quad ①$$

With $P_w - P_w\big|_{y=0} + \rho_w g y = \mu_w \frac{\partial v_w}{\partial y}$

$$\frac{\partial}{\partial x}\left(P_w - P_w\Big|_{y=0} + \rho_w g y \right) = \frac{\partial}{\partial x}\left(\mu_w \frac{\partial v_w}{\partial y} \right)$$

$$\therefore \frac{\partial P_w}{\partial x} - \frac{\partial P_w}{\partial x}\Big|_{y=0} = \mu_w \frac{\partial^2 v_w}{\partial x \partial y} \; ; \; \frac{\partial u_w}{\partial x} + \frac{\partial v_w}{\partial y} = 0$$

$$\frac{\partial P_w}{\partial x} - \frac{\partial P_w}{\partial x}\Big|_{y=0} = \mu_w \left(-\frac{\partial^2 u_w}{\partial x^2} \right)$$

$$\frac{\partial P_w}{\partial x} - \frac{\partial P_w}{\partial x}\Big|_{y=0} = -\mu_w \frac{\partial^2 u_w}{\partial x^2} \quad ③$$

$$\frac{\partial}{\partial x}\left(\frac{\partial u_w}{\partial x} + \frac{\partial v_w}{\partial y} \right) = \frac{\partial}{\partial x}(0)$$

$$\frac{\partial^2 u_w}{\partial x^2} + \frac{\partial^2 v_w}{\partial x \partial y} = 0, \quad \frac{\partial^2 v_w}{\partial x \partial y} = -\frac{\partial^2 u_w}{\partial x^2}$$

Incompressible Pipe Flow

So $$\frac{\partial P_w}{\partial x} - \frac{\partial P_w}{\partial x}\bigg|_{y=0} = -\mu_w \frac{\partial^2 u_w}{\partial x^2} \quad ③$$

$$\rho_w\left(\frac{\partial u_w}{\partial t} + u_w\frac{\partial u_w}{\partial x} + v_w\frac{\partial u_w}{\partial y}\right) = -\frac{\partial P_w}{\partial x} + \mu_w\frac{\partial^2 u_w}{\partial y^2} \quad ①$$

At wall, $y=0$,

$$\rho_w\left(\cancelto{0}{\frac{\partial u_w}{\partial t}\bigg|_{y=0}} + \cancelto{0}{u_w}\frac{\partial u_w}{\partial x} + \cancelto{0}{v_w}\frac{\partial u_w}{\partial y}\right) = -\frac{\partial P_w}{\partial x}\bigg|_{y=0} + \mu_w\frac{\partial^2 u_w}{\partial y^2}\bigg|_{y=0}$$

$u_w = 0$, no slip $\quad$ $v_w = 0$, no permeability

$u_w = 0$, no slip

$$0 = -\frac{\partial P_w}{\partial x}\bigg|_{y=0} + \mu_w\frac{\partial^2 u_w}{\partial y^2}\bigg|_{y=0}$$

$$\frac{\partial P_w}{\partial x}\bigg|_{y=0} = \mu_w\frac{\partial^2 u_w}{\partial y^2}\bigg|_{y=0}$$

$$\frac{\partial P_w}{\partial x} - \mu_w\frac{\partial^2 u_w}{\partial y^2}\bigg|_{y=0} = -\mu_w\frac{\partial^2 u_w}{\partial x^2}$$

$$\frac{\partial P_w}{\partial x} = \mu_w\frac{\partial^2 u}{\partial y^2} - \rho_w\left(\frac{\partial u_w}{\partial t} + u_w\frac{\partial u_w}{\partial x} + v_w\frac{\partial u_w}{\partial y}\right)$$

$\therefore$ Substitute,

$$\mu_w\frac{\partial^2 u}{\partial y^2} - \rho_w\left(\frac{\partial u_w}{\partial t} + u_w\frac{\partial u_w}{\partial x} + v_w\frac{\partial u_w}{\partial y}\right) - \mu_w\frac{\partial^2 u}{\partial y^2}\bigg|_{y=0} = -\mu_w\frac{\partial^2 u_w}{\partial x^2}$$

Incompressible Pipe Flow

So, $\mu_w \frac{\partial^2 u_w}{\partial y^2} - \rho_w \left(\frac{\partial u_w}{\partial t} + u_w \frac{\partial u_w}{\partial x} + v_w \frac{\partial u_w}{\partial y} \right) - \mu_w \frac{\partial^2 u}{\partial y^2}\Big|_{y=0} = -\mu_w \frac{\partial^2 u_w}{\partial x^2}$

$$\mu_w \frac{\partial^2 u_w}{\partial y^2} + \mu_w \frac{\partial^2 u_w}{\partial x^2} - \mu_w \frac{\partial^2 u}{\partial y^2}\Big|_{y=0} = \rho_w \left(\frac{\partial u_w}{\partial t} + u_w \frac{\partial u_w}{\partial x} + v_w \frac{\partial u_w}{\partial y} \right)$$

Nondimensionalize left-side of equation

$$\rightarrow \mu_w \frac{U_\infty}{h^2} \frac{\partial^2 u_w'}{\partial y'^2} + \mu_w \frac{U_\infty}{L^2} \frac{\partial^2 u_w'}{\partial x'^2} - \mu_w \frac{\partial^2 u}{\partial y^2}\Big|_{y=0} = \rho_w \left(\frac{\partial u_w}{\partial t} + u_w \frac{\partial u_w}{\partial x} + v_w \frac{\partial u_w}{\partial y} \right)$$

$$\mu_w \frac{U_\infty}{h^2} \left(\frac{\partial^2 u_w'}{\partial y'^2} + \underbrace{\cancel{\frac{h^2}{L^2} \frac{\partial^2 u_w'}{\partial x'^2}}}_{0,\, h \ll L} \right) - \mu_w \frac{\partial^2 u}{\partial y^2}\Big|_{y=0} = \rho_w \left(\frac{\partial u_w}{\partial t} + u_w \frac{\partial u_w}{\partial x} + v_w \frac{\partial u_w}{\partial y} \right)$$

$$\mu_w \frac{U_\infty}{h^2} \frac{\partial^2 u_w'}{\partial y'^2} - \mu_w \frac{\partial^2 u}{\partial y^2}\Big|_{y=0} = \rho_w \left(\frac{\partial u_w}{\partial t} + u_w \frac{\partial u_w}{\partial x} + v_w \frac{\partial u_w}{\partial y} \right)$$

$$\mu_w \frac{\partial^2 u_w}{\partial y^2} - \mu_w \frac{\partial^2 u}{\partial y^2}\Big|_{y=0} = \rho_w \left(\frac{\partial u_w}{\partial t} + u_w \frac{\partial u_w}{\partial x} + v_w \frac{\partial u_w}{\partial y} \right)$$

$$\rho_w \left(\frac{\partial u_w}{\partial t} + u_w \frac{\partial u_w}{\partial x} + v_w \frac{\partial u}{\partial y} \right) = \mu_w \frac{\partial^2 u_w}{\partial y^2} - \mu_w \frac{\partial^2 u_w}{\partial y^2}\Big|_{y=0} \quad ①$$

$$P_w - P_w\big|_{y=0} + \rho_w g y = \mu_w \frac{\partial v}{\partial y} \quad ②$$